

Dual Astable 555 Oscillators

The circuit diagram illustrates a Dual Astable 555 Oscillator. It features a central 7556 dual monostable timer IC (IC1) configured as two astable multivibrators. The left oscillator (PITCH1) uses a 10k resistor (R1), a 100nF capacitor (C1), and a 470 ohm resistor (R3) connected to a potentiometer (POT1). The right oscillator (PITCH2) uses a 10k resistor (R2), a 100nF capacitor (C2), and a 470 ohm resistor (R4) connected to a potentiometer (POT2). Both oscillators are powered by a 5V supply (VCC) and ground (GND). The output of the left oscillator is connected to a 1uF capacitor (C3) and a 1Meg resistor (R5) connected to a pushbutton (BUTTON). The output of the right oscillator is connected to a 1uF capacitor (C4) and a 1Meg resistor (R5) connected to a pushbutton (BUTTON). The circuit also includes a decoupling capacitor (C5) near the IC1.

Decoupling cap – place near IC1

Arrow on pots is backwards – it indicates counterclockwise rotation.

Divide Frequency by 2 (Sub-octave)

The image shows two circuit diagrams for 555 timer dividers and a decoupling capacitor.

IC2A and IC2B: These are 555 timer chips configured as frequency dividers. For IC2A, the clock input (pin 3, labeled CLK) is connected to PITCH1, and the reset pin (pin 4, labeled R) is connected to GND. The output (pin 1, labeled Q) is connected to SUB1, and the inverted output (pin 2, labeled Q̄) is connected to GND. The same configuration is shown for IC2B, with PITCH2 connected to the clock input (pin 11, labeled CLK) and SUB2 connected to the output (pin 13, labeled Q).

Decoupling capacitor C16: This is a .1uF capacitor connected between VCC and GND, placed near IC2C to decouple the power supply.

Mixer (Inverting Summer)

The diagram illustrates an inverting summer mixer circuit. It features four input channels, each consisting of a switch (S1, S2, S3, S4) controlled by a microcontroller (JP9, M015MDNS). The switches are connected to input signals PITCH1, SUB1, PITCH2, and SUB2. Each switch is also connected to a 10uF/25V capacitor (C6, C7, C8, C9, C10) and a 100k resistor (R6, R7, R8, R9, R10). The other end of the resistors is connected to the inverting input (pin 2) of the LM358KIC op-amp. The non-inverting input (pin 3) is connected to a voltage divider consisting of a 10k resistor (R11) and a 100k resistor (R10) connected to VCC and GND, respectively. The op-amp is powered by VCC (pin 8) and GND (pin 4). The output of the op-amp (pin 1) is connected to the output of the mixer.

Bridged-T Filter

The diagram illustrates a Bridged-T Filter circuit. It features an LM358 op-amp (IC3B) configured as a voltage follower. The non-inverting input (pin 5) is connected to a voltage divider consisting of two resistors (RV3B and RV3A) and two potentiometers (JP10, JP11, JP12, JP13). The output of the op-amp (pin 7) is connected to a feedback network consisting of two capacitors (C12 and C13) and a resistor (C11). The output of the filter is taken from the junction between C12 and C13. The circuit is powered by a 5V supply (pin 6) and ground (pin 7).

Output Amplifier

The diagram illustrates an output amplifier circuit. It features two LM358 op-amp chips, labeled IC4A and IC4B. The input signal, labeled VOLUME, is connected to a potentiometer (RV4) and then to the non-inverting input (pin 3) of IC4A. The inverting input (pin 2) of IC4A is connected to the non-inverting input (pin 5) of IC4B. The output of IC4A (pin 1) is connected to a 10uF/25V capacitor (C14), which is then connected to the AUDIO-JACK2KIT (J1). The output of IC4B (pin 7) is connected to a 10uF/25V capacitor (C15), which is then connected to the output of the AUDIO-JACK2KIT (J1). The output of the AUDIO-JACK2KIT (J1) is connected to two 01SMDNS components, labeled JP14 and JP15. The power supply (VCC) is connected to pin 8 of IC4A and pin 6 of IC4B. The ground (GND) is connected to pin 4 of IC4A and pin 5 of IC4B.

Battery, Power Switch & Trigger

The diagram illustrates a circuit for a battery, power switch, and trigger. The components and their connections are as follows:

- Battery:** A 9V battery (BAT1) is connected to the circuit. The positive terminal is connected to VCC, and the negative terminal is connected to GND.
- Power Switch:** A switch (S6) is connected to VCC. It is controlled by an SPDT switch (D3, 1n5819) and is connected to VCC.
- Trigger Switch:** A switch (S5) is connected to VCC. It is controlled by a momentary switch (S5) and is connected to VCC.
- Resistors:**
 - R12 (10k) and R13 (10k) are connected between VCC and GND.
 - R14 (1k) is connected between VCC and GND.
- Other Components:**
 - JP16 and JP17 are M015MDNS components.
 - VREF is a reference voltage point.

